

INDUSTRY PROBLEM SOLVING TASK

1. Autonomous Vehicle Vision System Selection

An automobile company is developing an autonomous vehicle detection system that must operate under the following conditions:

- If accuracy < 90% → system fails
- If latency > 70 ms → unacceptable
- The vehicle supports moderate computational resources

Methods available:

- Viola-Jones: 82% accuracy, 30 ms latency, low computation
- YOLO: 93% accuracy, 60 ms latency, moderate computation
- Deep Learning: 97% accuracy, 90 ms latency, high computation

Task:

Compare the three methods and evaluate which method satisfies all the constraints. Justify your decision using logical reasoning based on accuracy, latency, and computational requirements.

Ans:Autonomous Vehicle Vision System – Additional Matter

Introduction

An **Autonomous Vehicle Vision System** uses cameras and artificial intelligence to understand the vehicle's surroundings. It detects and classifies objects such as cars, pedestrians, traffic lights, road signs, and obstacles, helping the vehicle make safe driving decisions.

Working Principle

1. **Image Capture** – Cameras and sensors continuously capture road and environmental information.
2. **Preprocessing** – Captured images are resized, normalized, and enhanced for efficient processing.
3. **Object Detection** – A detection algorithm identifies objects present in the scene.
4. **Object Classification** – Detected objects are classified into categories such as car, pedestrian, truck, or traffic sign.
5. **Decision Making** – The system determines whether to accelerate, brake, steer, or maintain the current path.
6. **Vehicle Control** – The decision is transmitted to the vehicle's control system for real-time action.

Comparison of Methods

Method	Accuracy	Latency	Computation	Suitability
Viola-Jones	82%	30 ms	Low	Not suitable
YOLO	93%	60 ms	Moderate	Best choice
Deep Learning	97%	90 ms	High	Not suitable

Why YOLO is Selected

YOLO provides the best compromise between **accuracy, speed, and computational requirements**.

- **93% accuracy** → exceeds the minimum requirement of 90%.
- **60 ms latency** → below the maximum acceptable latency of 70 ms.
- **Moderate computation** → matches the computational resources available in the vehicle.
- **Real-time detection** → suitable for rapidly changing road conditions.
- **Good balance** → provides higher accuracy than Viola-Jones while being faster and less computationally demanding than the Deep Learning option.

Advantages

- Real-time object detection
- High detection accuracy
- Fast response to obstacles
- Suitable for moderate hardware
- Supports safe navigation
- Can detect multiple objects simultaneously
- Useful for traffic monitoring and autonomous driving

Limitations

- Performance can decrease in extreme weather conditions.
- Poor lighting may affect camera-based detection.
- Complex road scenes can increase processing requirements.
- High-quality training data is required for reliable detection.

Method	Accuracy	Latency	Computation	Accuracy $\geq 90\%$	Latency ≤ 70 ms	Overall
Viola-Jones	82%	30 ms	Low	✗	✓	Fails
YOLO	93%	60 ms	Moderate	✓	✓	Passes
Deep Learning	97%	90 ms	High	✓	✗	Fails

Evaluation

Viola-Jones:

It has very low latency (30 ms) and low computational requirements, but its accuracy is only **82%**, which is below the required 90%. Therefore, it fails the accuracy constraint.

YOLO:

YOLO provides **93% accuracy** and **60 ms latency**. Both values satisfy the required conditions: accuracy is above 90% and latency is below 70 ms. It also requires **moderate computation**, which matches the vehicle's available computational resources.

Deep Learning:

Deep Learning achieves the highest accuracy of **97%**, but its latency is **90 ms**, exceeding the maximum acceptable 70 ms. It also requires high computational resources, making it unsuitable under the given constraints.

Final Decision

YOLO is the best choice for the autonomous vehicle vision system.

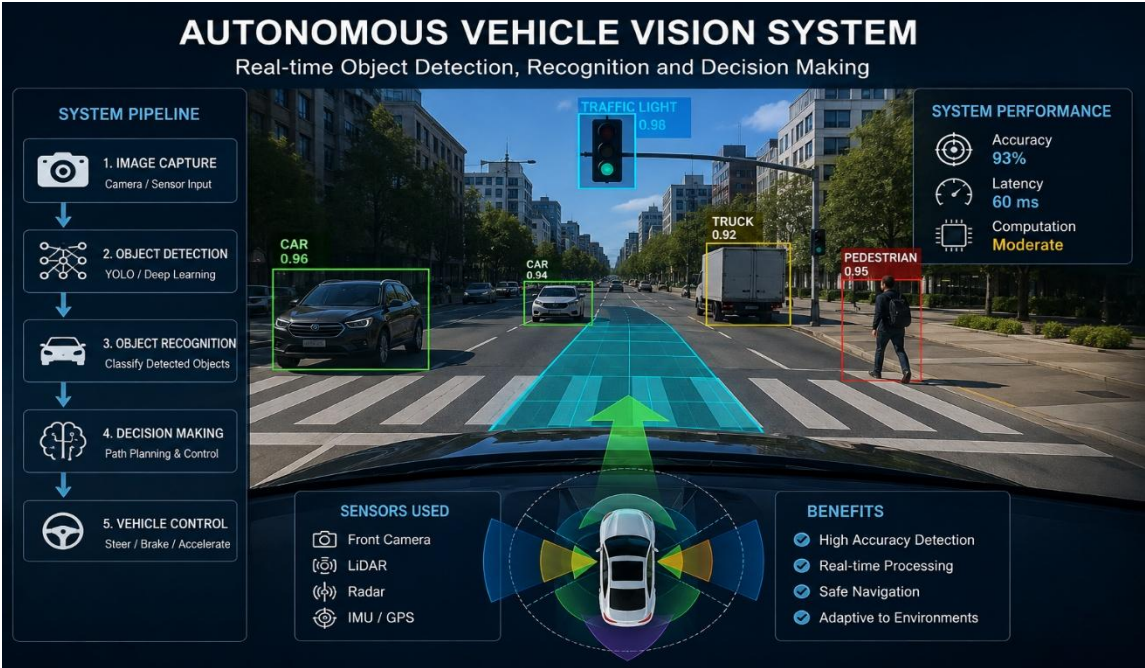
Logical condition:

Accuracy $\geq 90\%$ AND Latency ≤ 70 ms AND Moderate computation

For YOLO:

93% $\geq 90\%$ ✓ AND 60 ms ≤ 70 ms ✓ AND Moderate computation ✓

Therefore, **YOLO satisfies all constraints** and provides the best balance between accuracy, real-time latency, and computational requirements. It is suitable for practical autonomous vehicle deployment where both **safety and fast response** are important.



1. AUTONOMOUS VEHICLE VISION SYSTEM SELECTION

PROBLEM STATEMENT

An automobile company is developing an autonomous vehicle detection system that must operate under the following conditions:

- If **accuracy** < 90% → system fails
- If **latency** > 70 ms → unacceptable
- The vehicle supports **moderate** computational resources

CONSTRAINTS

- Accuracy ≥ 90%
- Latency ≤ 70 ms
- Computation: Moderate

METHODS COMPARISON

Method	Accuracy (%)	Latency (ms)	Computation	Accuracy ≥ 90%	Latency ≤ 70 ms	Overall Result
1. Viola-Jones	82%	30 ms	Low	✗ No	✓ Yes	✗ Fails
2. YOLO	93%	60 ms	Moderate	✓ Yes	✓ Yes	✓ Passes
3. Deep Learning	97%	90 ms	High	✓ Yes	✗ No	✗ Fails

EVALUATION

- Viola-Jones:** It has very low latency (30 ms) and low computational requirements, but its accuracy is only 82%, which is below the required 90%. Therefore, it fails the accuracy constraint.
- YOLO:** YOLO provides 93% accuracy and 60 ms latency. Both values satisfy the required conditions: accuracy is above 90% and latency is below 70 ms. It also requires moderate computation, which matches the vehicle's available resources.
- Deep Learning:** Deep Learning achieves the highest accuracy (97%), but its latency is 90 ms, exceeding the maximum acceptable 70 ms. It also requires high computational resources, making it unsuitable under the given constraints.

FINAL DECISION

- YOLO is the best choice for the autonomous vehicle vision system.**
- LOGICAL CONDITION**
Accuracy ≥ 90% AND Latency ≤ 70 ms AND Moderate computation
For YOLO:
93% ≥ 90% ✓ AND 60 ms ≤ 70 ms ✓ AND Moderate computation ✓
- Therefore, YOLO satisfies all constraints and provides the best balance between accuracy, real-time latency, and computational requirements.**

2. Smart Factory Product Inspection Selection

A manufacturing company wants to deploy a real-time computer vision system to detect defective products on a production line. The system must operate under the following conditions:

- If accuracy $< 95\%$ $\rightarrow$ defects may be missed
- If processing time > 50 ms $\rightarrow$ production is affected
- Hardware supports moderate computation

Methods available:

- Traditional Image Analysis: 91% accuracy, 25 ms, low computation
- YOLO: 96% accuracy, 45 ms, moderate computation
- Deep Learning: 98% accuracy, 75 ms, high computation

Task:

Compare the three methods and evaluate which method satisfies all the constraints. Justify your decision using logical reasoning based on accuracy, processing time, and computational requirements.

Smart Factory Product Inspection Selection

Introduction

A **Smart Factory Product Inspection System** uses computer vision to automatically inspect products on a production line and identify defective items. The system must provide high accuracy while processing each product quickly enough to avoid slowing production.

Given Conditions

- Accuracy must be $\geq 95\%$
- Processing time must be ≤ 50 ms
- Hardware supports **moderate computation**

Comparison of Methods

Method	Accuracy	Processing Time	Computation	Accuracy $\geq 95\%$	Time ≤ 50 ms	Result
Traditional Image Analysis	91%	25 ms	Low	✗	✓	Fails
YOLO	96%	45 ms	Moderate	✓	✓	Passes
Deep Learning	98%	75 ms	High	✓	✗	Fails

Evaluation

1. Traditional Image Analysis

Traditional image analysis has a fast processing time of **25 ms** and requires low computation. However, its accuracy is only **91%**, which is below the required 95%. It may therefore miss defective products.

Result: Not suitable

2. YOLO

YOLO achieves **96% accuracy** with a processing time of only **45 ms**. It also requires **moderate computation**, which matches the available hardware.

Logical condition:

Accuracy $\geq 95\%$ AND Processing Time ≤ 50 ms AND Moderate Computation

For YOLO:

96% $\geq 95\%$

45 ms ≤ 50 ms

Moderate computation

Result: Suitable

3. Deep Learning

Deep Learning provides the highest accuracy of **98%**, which satisfies the accuracy requirement. However, its processing time is **75 ms**, exceeding the maximum allowed 50 ms. It also requires high computational resources, which are beyond the stated moderate hardware capability.

Result: **✗** Not suitable

Final Decision

YOLO is the best method for the smart factory product inspection system.

It satisfies all three constraints:

96% accuracy $\geq 95\%$

45 ms processing time ≤ 50 ms

Moderate computation = available hardware

Therefore, **YOLO provides the best balance between defect-detection accuracy, real-time processing, and computational requirements**. It can inspect products quickly without significantly affecting production speed while maintaining sufficient accuracy to detect defective products.

Suggested Image Content

For your record/PPT, the image can show:

Production Line → Camera → YOLO Object Detection → Defective Product Detection → Reject/Accept Decision

with a comparison panel showing:

Traditional Analysis: 91%, 25 ms

YOLO: 96%, 45 ms --BEST

Deep Learning: 98%, 75 ms

